Problem 1. ***Pre-testing and uniformity problems.*** *Suppose you observe $X_1, \dots, X_n$ drawn IID $N(\mu, 1)$. The parameter space for μ is $(0, \infty)$. Since the parameter space is $(0, \infty)$, consider a researcher who reports the following confidence interval for μ:*

$$\widehat{CI} = \begin{cases} \bar{X}_n \pm 1.96/\sqrt{n} & \text{if } \bar{X}_n \geq 1.645/\sqrt{n}, \\ \emptyset, & \text{otherwise.} \end{cases}$$

The interpretation of this confidence interval is that the null hypothesis $\mu \leq 0$ is first tested at the 5% level. If the test rejects, the standard 95% confidence interval is reported; otherwise the empty set is reported. Let P_μ denote probabilities when μ is the true parameter.

(a) Show that the $\widehat{CI}$ is valid in a point-wise sense. That is, for all $\mu \in (0, \infty)$,

$$\lim_{n\to\infty} P_\mu(\mu \in \widehat{CI}) = 0.95.$$

(b) Now consider a sequence of DGPs indexed by sample size n and some $c > 0$, in which $\mu = c/\sqrt{n}$. Find

$$\lim_{n\to\infty} P_{c/\sqrt{n}}(c/\sqrt{n} \in \widehat{CI}).$$

(c) Based on the answer to (b), determine whether $\widehat{CI}$ is uniformly valid. That is, decide whether

$$\lim_{n\to\infty} \inf_{\mu\in(0,\infty)} P_\mu(\mu \in \widehat{CI}) \geq 0.95$$

holds. (Bonus points if the left-hand side is calculated analytically.)

Solution.

(a) Fix $\mu > 0$. The pre-test threshold satisfies $1.645/\sqrt{n} \to 0$ as $n \to \infty$.

By the law of large numbers, $\bar{X}_n \xrightarrow{p} \mu > 0$. Equivalently, by the central limit theorem,

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} Z \sim N(0, 1)$$

and $\sqrt{n}\mu \to \infty$. The standardized pre-test threshold is

$$\sqrt{n}\Big(\frac{1.645}{\sqrt{n}} - \mu\Big) = 1.645 - \sqrt{n}\mu \to -\infty.$$

Therefore

$$P_\mu\Big(\bar{X}_n < \frac{1.645}{\sqrt{n}}\Big) \to 0,$$

which means $P_\mu(\widehat{CI} = \emptyset) \to 0$.

On the event that the pre-test is passed (which has probability tending to 1), $\widehat{CI}$ coincides exactly with the usual symmetric 95% confidence interval. Moreover,

$$P_\mu\Big(|\sqrt{n}(\bar{X}_n - \mu)| \leq 1.96 \mid \text{pre-test passed}\Big) \to 0.95$$

by the central limit theorem together with the fact that the conditioning event has probability tending to 1. The unconditional coverage therefore converges to 0.95.

(b) Let $\mu_n = c/\sqrt{n}$ with $c > 0$. Define

$$W_n = \sqrt{n}(\bar{X}_n - \mu_n) \xrightarrow{d} Z \sim N(0,1).$$

The pre-test is passed if and only if $\bar{X}_n \geq 1.645/\sqrt{n}$, or equivalently

$$W_n \geq 1.645 - c.$$

The interval covers μ_n if and only if the pre-test is passed *and* $|\bar{X}_n - \mu_n| \leq 1.96/\sqrt{n}$, i.e.,

$$|W_n| \leq 1.96.$$

Therefore the limiting coverage probability is

$$P\big(\max(1.645 - c, -1.96) \leq Z \leq 1.96\big) = \Phi(1.96) - \Phi\big(\max(1.645 - c, -1.96)\big).$$

(c) The confidence interval $\widehat{CI}$ is *not* uniformly valid. From part (b), as $c \to 0^+$ the limiting coverage converges to

$$\Phi(1.96) - \Phi(1.645) \approx 0.975 - 0.95 = 0.025.$$

For every fixed $\mu > 0$ the coverage tends to 0.95 (by part (a)). As $\mu \downarrow 0$ the asymptotic coverage therefore approaches 0.025. Consequently

$$\lim_{n\to\infty} \inf_{\mu>0} P_\mu(\mu \in \widehat{CI}) = 0.025 < 0.95.$$

(The infimum is attained in the limit as $\mu \downarrow 0$.)

□

Problem 2. *Wold decomposition.* *The Wold decomposition asserts that a stationary process $X = \{X_t : t \in \mathbb{Z}\}$ with mean zero and finite variance can generally be written as*

$$X_t = \sum_{i=0}^{\infty} \gamma_i \varepsilon_{t-i}, \tag{1}$$

where $\gamma_0 = 1$, $\sum_{i=0}^{\infty} \gamma_i^2 < \infty$, $E[\varepsilon_t] = 0$, and $E[\varepsilon_t \varepsilon_s] = 0$ for $t \neq s$, with

$$\varepsilon_t = X_t - \hat{X}_t,$$

where $\hat{X}_t$ is the best linear predictor of X_t given $X_{t-1}, X_{t-2}, \ldots$. That is, $\hat{X}_t$ solves

$$\min_{Y_t \in \mathrm{Y}_t} E[(X_t - Y_t)^2],$$

where Y_t denotes all random variables of the form $Y_t = \sum_{i=1}^{\infty} a_i X_{t-i}$ with $E[Y_t^2] < \infty$. As the ε_t are uncorrelated, γ_i is the population regression coefficient of X_t on ε_{t-i}:

$$\gamma_i = \frac{E[X_t \varepsilon_{t-i}]}{E[\varepsilon_t^2]},$$

provided $E[\varepsilon_t^2] > 0$. Any process X that can be written as (1) with independent ε_t is said to be a moving average process. Independence of the ε_t allows the γ_i to be interpreted as impulse response functions.

(a) *The projection theorem implies $E[(X_t - \hat{X}_t)Y_t] = 0$ for all $Y_t \in Y_t$. Use this to explain why $E[\varepsilon_t \varepsilon_s] = 0$ for $t \neq s$. (Start by showing $E[\varepsilon_{t+1}\varepsilon_t] = 0$.)*

(b) *Consider the AR(1) model $X_t = \rho X_{t-1} + e_t$, where $|\rho| < 1$ and e_t are IID with mean 0 and variance σ^2.*

(i) *Show $\hat{X}_t = \rho X_{t-1}$.*

(ii) *Derive the Wold decomposition of X. Give expressions for the γ_i and ε_t.*

(iii) *Is X a moving average process?*

(c) *Let $\{X_t : t \in \mathbb{Z}\}$ be a two-state Markov chain with state-space $\{-1, 1\}$ and symmetric, time homogeneous transition probabilities. That is, each X_t takes values in $\{-1, 1\}$. Moreover, given X_{t-1},*

$$X_t = \begin{cases} X_{t-1} & \text{with probability } 1-p, \\ -X_{t-1} & \text{with probability } p, \end{cases}$$

where $p \in (0, 1)$.

(i) *Show that the stationary distribution is $P(X_t = 1) = P(X_t = -1) = \frac{1}{2}$.*

(ii) *Calculate $E[X_t|\mathcal{F}_{t-1}]$.*

(iii) *Use the previous answer to find $\hat{X}_t$.*

(iv) *Derive the Wold decomposition of X. Give expressions for the γ_i and ε_t.*

(v) *Is X a moving average process when $p \neq \frac{1}{2}$? Explain.*

(vi) *Is X a moving average process when $p = \frac{1}{2}$? Explain.*

Solution.

(a) Let $\mathcal{H}_{t-1} = \overline{\text{span}}\{X_{t-1}, X_{t-2}, \dots\}$ denote the closed linear span of the process history. By the definition of the best linear predictor, the prediction error $\varepsilon_t = X_t - \hat{X}_t$ is orthogonal to this entire subspace. That is, $E[\varepsilon_t Y] = 0$ for all $Y \in \mathcal{H}_{t-1}$.

Consider any $s < t$. The error term ε_s is a linear combination of X_s and its past values, implying $\varepsilon_s \in \mathcal{H}_s$. Since time flows forward, the history at time s is contained within the history at time $t-1$ (i.e., $\mathcal{H}_s \subseteq \mathcal{H}_{t-1}$). Consequently, ε_s is an element of the space $\mathcal{H}_{t-1}$. Since ε_t is orthogonal to every element in $\mathcal{H}_{t-1}$, it follows that $E[\varepsilon_t \varepsilon_s] = 0$.

(b) (i) The process possesses the Markov property and the conditional expectation satisfies $E[X_t|X_{t-1}] = \rho X_{t-1}$. Because further lags contribute no additional information, the best linear predictor based on the infinite past is $\hat{X}_t = \rho X_{t-1}$.

(ii) Repeated substitution of the AR(1) recursion produces the MA(∞) representation

$$X_t = \sum_{j=0}^{\infty} \rho^j e_{t-j}, \qquad \gamma_i = \rho^i.$$

Equivalently, $\varepsilon_t = e_t$ and

$$\gamma_i = \frac{E[X_t \varepsilon_{t-i}]}{E[\varepsilon_t^2]} = \frac{E[X_t e_{t-i}]}{\sigma^2} = \rho^i.$$

(iii) The innovations $\{e_t\}$ are independent and identically distributed, so the representation is a moving average process.

(c) (i) Let $\pi = (\pi_1, 1 - \pi_1)$. Stationarity requires

$$\pi_1 = \pi_1(1-p) + (1-\pi_1)p,$$

which yields $\pi_1 = 1/2$.

(ii) The Markov property gives

$$E[X_t|\mathcal{F}_{t-1}] = E[X_t|X_{t-1}] = (1-p)X_{t-1} + p(-X_{t-1}) = (1-2p)X_{t-1}.$$

(iii) Because the conditional expectation is a linear function of X_{t-1}, the best linear predictor equals $\hat{X}_t = (1-2p)X_{t-1}$.

(iv) Set $\varepsilon_t = X_t - (1-2p)X_{t-1}$. The autocovariance function is $\gamma(k) = (1-2p)^{|k|}$ with stationary variance $\gamma(0) = 1$. Then

$$E[\varepsilon_t^2] = 1 - (1-2p)^2 = 4p(1-p).$$

For $i \geq 1$,

$$\gamma_i = \frac{\operatorname{Cov}(X_t, \varepsilon_{t-i})}{E[\varepsilon_t^2]} = \frac{\operatorname{Cov}(X_t, X_{t-i}) - (1-2p)\operatorname{Cov}(X_t, X_{t-i-1})}{4p(1-p)}$$
$$= \frac{(1-2p)^i - (1-2p)^{i+2}}{4p(1-p)} = (1-2p)^i.$$

and $\gamma_0 = 1$. The Wold representation is therefore

$$X_t = \sum_{i=0}^{\infty} (1-2p)^i \varepsilon_{t-i}.$$

(v) No. The innovations $\{\varepsilon_t\}$ remain uncorrelated but fail to be independent when $p \neq 1/2$. Each ε_t is a deterministic function of X_{t-1} and the current transition, while X_{t-1} depends on $\{\varepsilon_s : s < t\}$. The mapping from past shocks to future values is nonlinear; consequently the coefficients γ_i do not admit the interpretation of impulse responses.

(vi) Yes. When $p = 1/2$ the predictor $\hat{X}_t$ equals zero and $\varepsilon_t = X_t$. The sequence is i.i.d. taking values ± 1 with equal probability, and the innovations are independent.

□